

DOAC Things to Note

By Li Suan

DOAC is High Alert Med!

If taking wrongly,
Bleeding → Death!

Your role is important in
educating patient

Please read these slides beforehand

All files in this folder:

- 1) W:\Farmasi\Garis Panduan & Rujukan\2. Clinical\2. Antiplatelet & Anticoagulant\DOAC
- 2) P:\Pharmacy\COVID-19 & AMS Guides - read the **Pharmacy HSgB COVID-19 Guide**

ANTICOAGULANT CONVERSION CHART

Newsletter by: PRIC Hospital Sg Buloh, Pua Fu Siang & Ooi Yu Xin

To From	Warfarin	Dabigatran	Rivaroxaban	Apixaban	Parenteral
LMWH	Overlapped at least 5 days OR until 2 consecutive days with INR in therapeutic range. ^{1,2}	Start DOAC at the next scheduled dose of LMWH/Fondaparinux. ^{1,3,4}			Treatment: ² Stop LMWH/Fondaparinux; start IV UFH 1-2 hours before next dose of LMWH/Fondaparinux. Prophylactic: ⁵ Stop LMWH/Fondaparinux; start SC UFH at the next scheduled dose of LMWH/Fondaparinux.
Fondaparinux		Start DOAC at time of discontinuation of continuous UFH. ^{1,3}			Stop continuous UFH; start LMWH/ Fondaparinux within 1 hour. ²
Continuous UFH					
Warfarin		Stop Warfarin; start Dabigatran when INR < 2.0. ^{1,3,6}	AF: ⁵ Stop Warfarin; start Rivaroxaban when INR ≤ 3.0. DVT or PE: ⁵ Stop Warfarin; start Rivaroxaban when INR ≤ 2.5.	Stop Warfarin; start Apixaban when INR < 2.0. ^{1,3,4}	Stop Warfarin; start parenteral anticoagulant when INR < 2.0. ⁷
Dabigatran	Option 1: Overlapping ⁶ Start Warfarin, then if: CrCl ≥ 50ml/min: Stop Dabigatran 3 days later CrCl 30-50ml/min: Stop Dabigatran 2 days later CrCl 15-30ml/min: Stop Dabigatran 1 day later CrCl <15ml/min: Dabigatran not recommended. ²	Stop current DOAC; start new DOAC at the next scheduled dose of previous DOAC. ⁶			CrCl ≥ 30 ml/min: ⁷ Stop Dabigatran; start parenteral agent 12 hours after last Dabigatran. CrCl < 30 ml/min: ⁷ Stop Dabigatran; start parenteral agent 24 hours after last Dabigatran. (Note: Fondaparinux contraindicated)
	Option 2: Bridging ⁷ Stop Dabigatran; start Warfarin with parenteral agent at next scheduled dose of Dabigatran. Cont. parenteral agent until INR in therapeutic range.				
Rivaroxaban/ Apixaban	Option 1: Overlapping ^{2,6} Start Warfarin; stop DOAC when INR in therapeutic range. Option 2: Bridging ^{2,6} Stop DOAC; start Warfarin with parenteral agent at next scheduled dose of DOAC. Cont. parenteral agent until INR in therapeutic range.				Stop Rivaroxaban/Apixaban; start parenteral agent at the next scheduled dose of Rivaroxaban/Apixaban. ^{1,7}

Only for internal circulation. For further enquiries, kindly contact ext 4126. Revised 21 July 2020.

References

1. CPG Prevention and Treatment of Venous Thromboembolism. 2013; 1-168.
2. Lexicomp® Drug Information handbook. 27th edition. Wolters Kluwer; 2018.
3. Anticoagulation MTAC (AC-MTAC) Protocol. Pharmaceutical Services Program 2020; 2:1-86.
4. Lofgren B. How to Convert Between Anticoagulants. Pharmacy Times. 2015.
5. Gonzalez C, Franco-Martinez C. Transitioning Between Anticoagulants. University Health System. Jan 2014.
6. Leung LK, Mannucci PM, Tinnauer JS. Direct oral anticoagulants (DOACs) and parenteral direct-acting anticoagulants: Dosing and adverse effects. UpToDate 2020.
7. Switching To and From Various Anticoagulants. Minneapolis Heart Institute®, Abbott Northwestern Hospital. Aug 2016.

When receive DOAC counseling referral (Pre-counseling screening)

1. Patient/carer agreed for DOAC?
2. Indication?
 - AF vs DVT dosing are different!
3. METHOD of SUPPLY?
 - Self purchase/Kuota HSB/Pesara
4. SCREENING CHECKLIST
 - Transient medical condition – Transaminitis
5. For **DVT/PE treatment**, patient may be on Clexane/Arixtra before switching to DOAC. Ask Dr when **planned to start DOAC**? Remind Dr to **OFF Clexane/Arixtra in system when starting DOAC** and **ORDER DOAC in eHIS**
6. If self purchase, when will patient get the DOAC stock?
 - Dabigatran: check with ward/note how many doses of enoxaparin/fondaparinux given in ward (**treatment dose** only)
 - Apixaban/Rivaroxaban – has own lead-in dose
 - Switching from any other anticoagulant? Refer DIS Anticoagulant Conversion Chart
1. For DVT/PE treatment - how to bridge warfarin and parenteral anticoagulant

Case study A

You received call from Dr to refer for DOAC counseling for Patient A

Q1: What do you need to ask the Dr?

1. Asking the right question

Q1: What do you need to ask the Dr?

- a. Patient/carer agreed for DOAC?
- b. Patient details (SB, name)
- c. Indication
- d. For screening: Age, BW, BMI, RP, Liver Profile
- e. Which DOAC?
- f. Dose of DOAC – DON'T make assumptions!
- g. Selfpurchase/Kuota HSB/Pesara
- h. Is the medicine with patient already? When to start?
- i. Counsel patient or carer?
- j. Check for potential drug interaction



DOAC Guideline in AF & VTE

Prepared by PRIC HSgB.

For internal circulation only. To consult physician if needed.

PLEASE REFER TO THE UPDATED GUIDE

CHECKLIST

INDICATION: STROKE
PREVENTION IN AF (SPAF)

INDICATION: VTE
**Not applicable for prevention of
VTE in patient undergoing hip or
knee replacement surgery*

*Duration for treatment: at least
3 months*

RENAL FUNCTION

Usual dose
150mg BD

Dosing Adjustment^[1,4]

- 110mg BD if fulfill any of the following risk factors:

- i) Patients aged ≥80 years
- ii) Concomitant use of Verapamil / Amiodarone

**Please refer to bottom of the table for further dosing consideration*

Treatment / Prevention^[1,4]

PARENTERAL ANTICOAGULANT for at least 5 days
THEN

150mg BD

OR

110mg BD if fulfill any of the following risk factors

- i) Patients aged ≥80 years
- ii) Concomitant use of Verapamil / Amiodarone
- iii) High bleeding risk AND CrCl: 30-50 ml/min

**Please refer to bottom of the table for further dosing consideration*

<30 ml/min: Contraindicated^[1,4]

5mg BD

Dosing Adjustment^[2]

- 2.5mg BD if fulfill TWO of the following criteria:

- i) Patients aged ≥80 years
- ii) Body weight ≤60kg
- iii) Serum Creatinine ≥133 umol/L

- 2.5mg BD if CrCl 15-29ml/min

Treatment^[2]

10mg BD for 7 days

THEN

5mg BD

Prevention of Recurrent VTE^[2]

2.5mg BD (after at least 6 months of therapeutic dose
of anticoagulant)

<15 ml/min : Contraindicated
15 – 29 ml/min : Use with caution^[2]

RIVAROXABAN

Usual Dose^[3]
20mg OD

Dosing Adjustment^[3]

- 15 mg OD if CrCl 15-49 ml/min

Treatment^[3]

15mg BD for 21 days

THEN

20mg once daily

Prevention of Recurrent VTE^[3]

10 mg OD (after at least 6 months of therapeutic
dose of anticoagulant)

<15 ml/min : Contraindicated
15 – 29 ml/min : Use with caution^[3]

*** FOR DABIGATRAN: May consider adjustment to 110mg BD based on individual thromboembolic and bleeding risk assessment for the following groups:**

i) Age: 75-80 years, ii) Moderate renal impairment (30 – 50ml/min), iii) Hx of gastritis, esophagitis or gastroesophageal reflux, iv) Patients with increased risk of bleeding

References

1. Pradaxa [Product Insert]. Ingelheim Rhein: Boehringer Ingelheim; 07 February 2020.
2. Eliquis [Product Insert]. New York: Pfizer; 03 September 2020.
3. Xarelto [Product Insert]. Leverkusen: Janssen Pharmaceuticals; March 2020.
4. (2020). Anticoagulant MTAC (AC-MTAC) Protocol. 2nd Ed. Selangor: Ministry of Health Malaysia.
5. Chen, A., Stecker, E., & Warden, B. A. (2020). Direct Oral Anticoagulant Use: A Practical Guide to Common Clinical Challenges. *Journal of the American Heart Association*, 1-18.
6. Rivaroxaban—Once daily, oral, direct factor Xa inhibition Compared with vitamin K antagonist for prevention of stroke and Embolism Trial in Atrial Fibrillation: Rationale and Design of the ROCKET AF study. (2010). *American Heart Journal*, 159(3). <https://doi.org/10.1016/j.ahj.2009.11.023>

Revised May 2021

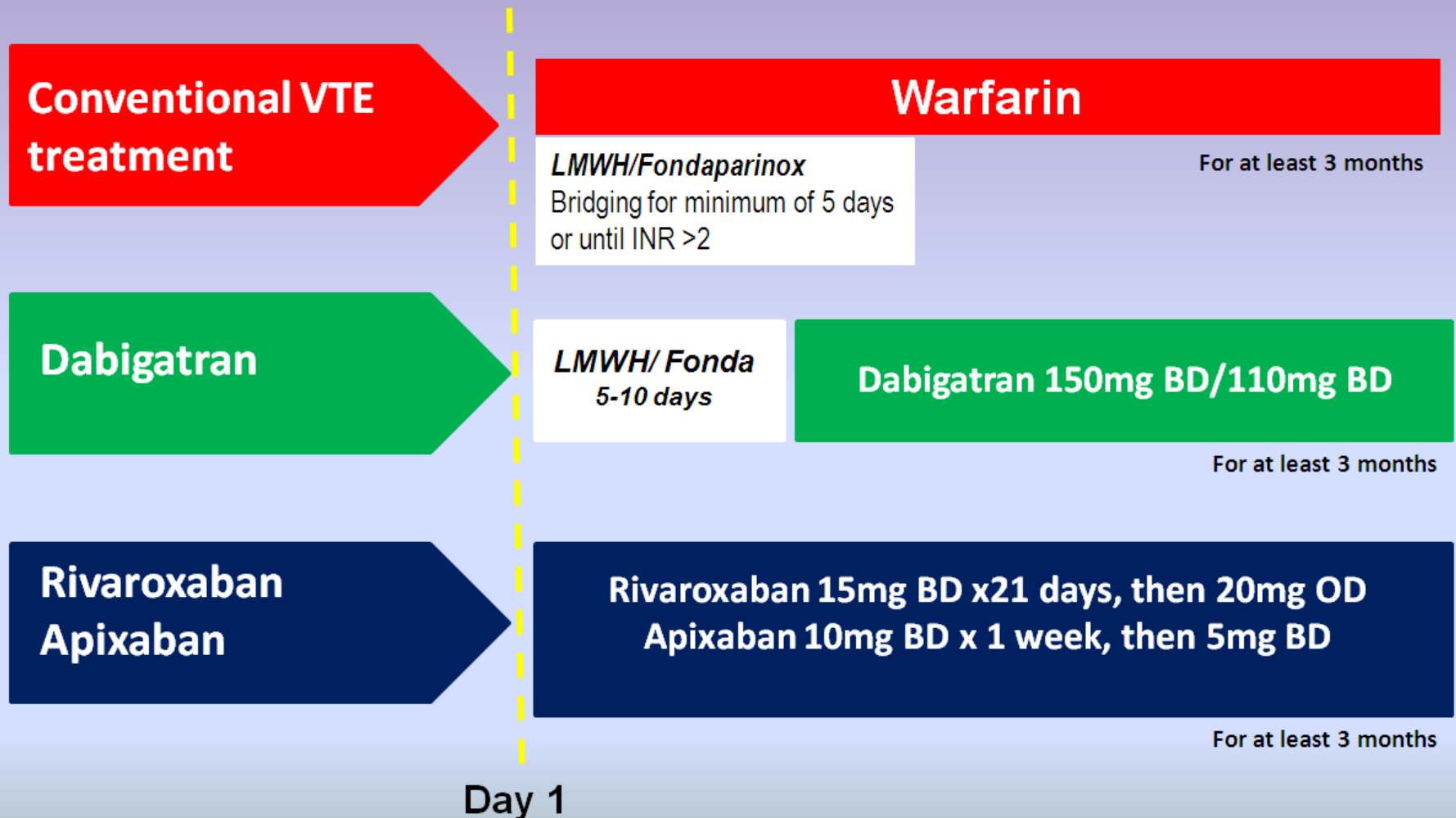


2. When to start?

Dr informed that patient agreed for T. Dabigatran for PE treatment . After you have screened against the DOAC checklist, you found that patient has no contraindication for Dabigatran. What is the next thing you need to ask?

- Current no of day/doses (**treatment dose**) on Enoxaparin/Fondaparinux
- When to start Dabigatran

Initiation of DOAC in acute VTE



ANTICOAGULANT CONVERSION CHART

Newsletter by: PRIC Hospital Sg Buloh, Pua Fu Siang & Ooi Yu Xin

To From	Warfarin	Dabigatran	Rivaroxaban	Apixaban	Parenteral
LMWH	Overlapped at least 5 days <u>OR</u> until 2 consecutive days with INR in therapeutic range. ^{1,2}	Start DOAC at the next scheduled dose of LMWH/Fondaparinux. ^{1,3,4}			Treatment: ² Stop LMWH/Fondaparinux; start IV UFH 1-2 hours before next dose of LMWH/Fondaparinux. Prophylactic: ⁵ Stop LMWH/Fondaparinux; start SC UFH at the next scheduled dose of LMWH/Fondaparinux.
Fondaparinux		Start DOAC at time of discontinuation of continuous UFH. ^{1,3}			Stop continuous UFH; start LMWH/ Fondaparinux within 1 hour. ²
Continuous UFH		Start DOAC at time of discontinuation of continuous UFH. ^{1,3}			Stop continuous UFH; start LMWH/ Fondaparinux within 1 hour. ²
Warfarin		Stop Warfarin; start Dabigatran when INR < 2.0. ^{1,3,6}	AF: ⁵ Stop Warfarin; start Rivaroxaban when INR ≤ 3.0. DVT or PE: ⁵ Stop Warfarin; start Rivaroxaban when INR ≤ 2.5.	Stop Warfarin; start Apixaban when INR < 2.0. ^{1,3,4}	Stop Warfarin; start parenteral anticoagulant when INR < 2.0. ⁷
Dabigatran	Option 1: Overlapping ⁶ Start Warfarin, then if: CrCl ≥ 50ml/min: Stop Dabigatran 3 days later CrCl 30-50ml/min: Stop Dabigatran 2 days later CrCl 15-30ml/min: Stop Dabigatran 1 day later CrCl <15ml/min: Dabigatran not recommended. ²	Stop current DOAC; start new DOAC at the next scheduled dose of previous DOAC. ⁶			CrCl ≥ 30 ml/min: ⁷ Stop Dabigatran; start parenteral agent 12 hours after last Dabigatran. CrCl < 30 ml/min: ⁷ Stop Dabigatran; start parenteral agent 24 hours after last Dabigatran. (Note: Fondaparinux contraindicated)
	Option 2: Bridging ⁷ Stop Dabigatran; start Warfarin with parenteral agent at next scheduled dose of Dabigatran. Cont. parenteral agent until INR in therapeutic range.				
Rivaroxaban/ Apixaban	Option 1: Overlapping ^{2,6} Start Warfarin; stop DOAC when INR in therapeutic range. Option 2: Bridging ^{2,6} Stop DOAC; start Warfarin with parenteral agent at next scheduled dose of DOAC. Cont. parenteral agent until INR in therapeutic range.				Stop Rivaroxaban/Apixaban; start parenteral agent at the next scheduled dose of Rivaroxaban/Apixaban. ^{1,7}

Only for internal circulation. For further enquiries, kindly contact ext 4126. Revised 21 July 2020.

References

1. CPG Prevention and Treatment of Venous Thromboembolism. 2013; 1-168.
2. Lexicomp® Drug Information handbook. 27th edition. Wolters Kluwer; 2018.
3. Anticoagulation MTAC (AC-MTAC) Protocol. Pharmaceutical Services Program 2020; 2:1-86.
4. Lofgren B. How to Convert Between Anticoagulants. Pharmacy Times. 2015.
5. Gonzalez C, Franco-Martinez C. Transitioning Between Anticoagulants. University Health System. Jan 2014.
6. Leung LK, Mannucci PM, Tinnauer JS. Direct oral anticoagulants (DOACs) and parenteral direct-acting anticoagulants: Dosing and adverse effects. UpToDate 2020.
7. Switching To and From Various Anticoagulants. Minneapolis Heart Institute®, Abbott Northwestern Hospital. Aug 2016.

2. When to start?

- Patient A is currently on Day 5 S/C Clexane 60mg BD, already injected today's morning dose. When is the correct time to start T. Dabigatran?
 - a) Can start T. Dabigatran anytime today
 - b) Start T. Dabigatran tonight at scheduled dose of Clexane
 - c) Start T. Dabigatran tomorrow morning on the scheduled dose of Clexane

To supply with leaflet

- Note:
ONLY FOR PE/DVT, NOT AF
- Write the starting date.
If Self-purchase:
 - Make sure stock already with patient before counseling
 - Check that patient bought the correct DOAC and strength

MAKLUMAT PENTING!

PANDUAN PENGAMBILAN UBAT ANTIKOAGULAN UNTUK



INDIKASI: PEMBEKUAN DARAH DALAM SALURAN DARAH PARU-PARU ATAU KAKI SAHAJA

Nama Pesakit: _____ SB No: _____ Wad: _____

DOS PERMULAAN UBAT ANTIKOAGULAN:

- ☒ Tablet Apixaban 5mg ⇒ DUA biji setiap 12 jam untuk 7 hari
- ☐ Tablet Rivaroxaban 15mg ⇒ SATU biji setiap 12 jam untuk 21 hari
- ☐ Capsule Dabigatran 110mg ⇒ SATU biji setiap 12 jam sehingga habis tempoh rawatan
- ☐ Capsule Dabigatran 150mg ⇒ SATU biji setiap 12 jam sehingga habis tempoh rawatan

TARIKH MULA	7/6/21	SEHINGGA	13/6/21
-------------	--------	----------	---------

KEMUDIAN, SAMBUNG MAKAN



DOS SAMBUNGAN UBAT ANTIKOAGULAN:

- ☒ Tablet Apixaban 5mg ⇒ SATU biji setiap 12 jam sehingga habis tempoh rawatan
- ☐ Tablet Rivaroxaban 20mg ⇒ SATU biji 1 kali sehari sehingga habis tempoh rawatan

TARIKH MULA	14/6/21	TARIKH DIJANGKA BERHENTI	7/9/21
-------------	---------	--------------------------	--------

VERY IMPORTANT! If you didn't tell them when to switch to maintenance dose, they may continue taking loading dose for 3 months and ended up with internal bleeding!

UBAT ANTIKOAGULAN PERLU DIAMBIL UNTUK TEMPOH SEKURANG-KURANGNYA 3 BULAN.
JANGAN BERHENTI MENGAMBIL UBAT INI TANPA BERBINCANG DENGAN DOKTOR ANDA.
UNTUK MAKLUMAN LANJUT DAN SEBARANG PERTANYAAN, SILA HUBUNGI JABATAN FARMASI HOSPITAL SUNGAI BULOH.

Important Points to tell patient

- When to start taking DOAC (PE/DVT Treatment)
 - Include the date in your DOAC counseling notes
- What dose and any lead-in dose?
- **Duration of treatment**
 - AF: lifelong
 - DVT/PE: 3-6months/ lifelong
- Ask patient to show the physical DOAC name & strength bought by pt
- Self-purchase or supplied by hospital
 - Ask Dr to order in system when starting
- All other counseling points

Raised LFT/Transaminitis

- Not recommended when patient's LFT are raised >2 ULN:
- Discuss with Dr, to wait until LFT <2 ULN and inform to refer for counseling again later on. Document in system as Pharmacist Note

SCREENING OF DOAC

- **COMPULSORY** to be screened by pharmacist before counsel by PRP
- You should also do your own screening !!
- Remember to include the name of the PF who screened in your counseling notes